



## Viticulture

### Lesson Aim

Describe the nature and scope of the Viticulture industry throughout the world.

### Global Viticulture

Some of the most recognisable regions in the world are those famous for their viticulture. France has districts with long histories of high-quality grape production and winemaking, such as the Champagne, Bordeaux, Burgundy, and Beaujolais districts. Winemakers in these areas have contributed greatly to the art and science of viticulture and the products that have emanated from France over the centuries have been applauded worldwide. Italians are equally proud of the contribution of wine-producing regions such as Tuscany. Germany's Rhine Valley and Mosel regions have an impressive heritage, as do districts in Spain and Portugal.

In the 21st century wines produced from much newer viticultural countries are also making a big impact on world markets. Examples include the Napa Valley in the United States and southern hemisphere nations such as Argentina, Chile, New Zealand and Australia. With a strong influence from European winemaking emigrants, the unique Australian soil and climatic conditions have favoured high-quality winemaking in regions such as the Hunter, the Barossa and Yarra Valleys as well as the west's Margaret River and southern Victoria's Mornington Peninsula. Countries that have only been producing wine for a mere century or two have relied on the science of Oenology for improved production techniques in place of "traditional" methods and inherited family secrets.

Suitable regions for good-quality grape production are determined more by climatic similarities than geographic location. Regions that have mean annual temperatures between 10 and 20 degrees Celsius are the most conducive for quality wine production. World distribution of viticulture is bounded by the 50° line of latitude, both north and south of the equator. However, even within these general parameters, grape vines are not suited to places where leaves do not fall from the vines over winter (due to warmth) or where winters are severe and summers are short.

Assessing regional suitability to grape production is not absolute. Variations in local climate caused by topographical characteristics can greatly affect the feasibility of production. For example, elevated areas in warm climate regions may yield the cooler temperatures to produce good quality winemaking grapes. Several parameters are commonly used for assessing growing conditions. Degree Days and Latitude-Temperature Index (LTI) are two such measures. A region with a higher latitude may have cooler mid-summer temperatures but may not be inhibited from good production when offset by a long growing season. The Bordeaux region of France and areas of Washington State in the USA may fall into this category.



In the UK, the most successful grape growing regions are located in the south and west. However, other regions could produce good crops if the viticulturist is equipped with suitable knowledge and a strong desire to grow good vines. Possibly, British production has historically been curtailed by the availability of good grapes and excellent wine from 'The Continent.' Small commercial viticulturists and informed home enthusiasts may find success where others have failed. Inadequate sugar development and disease problems have been stumbling blocks in the past. Appropriate site selection and preparation and good varietal choices can overcome many of these problems.

Indoor vine production is also viable in the UK and other areas with similar climates. Well-conditioned and managed soils can alleviate many disease problems and the environmental control offered by glasshouses can extend the opportunity for sufficient ripening. Specific production techniques are required for indoor production, particularly in regards to vine pruning and support. Container-grown vines can allow the grower to move plants indoors or outdoors to take advantage of favourable conditions.

Although it is believed that viticulture began in Asia, the Mediterranean region has probably had the most influence on the structure of modern vine growing and wine production. Early Greek growers used the single post method of vine training, in contrast to trellising. With the colonisation of the 'New World' came the spread of viticulture to the Americas, South Pacific, and even southern Africa.



Warmer climates are more suitable for producing dried fruit than high-quality wines. The best quality wines are generally from cooler climates. While not totally understood, it is thought that slower ripening and production of sugars in a cooler climate is a significant factor. If cold temperatures afflict a region, the vines may be damaged by frosts and freezes. Additionally, shortened summers give insufficient time for grapes to develop a full complement of sugars and flavours. Conversely, the rapid ripening of fruit in hot climates may also lead to undeveloped, inferior grapes not suitable for quality wine production.

## Some Major Winegrowing Areas Around the World

Grapes are indigenous to temperate climates, most likely in the Caucasus region. The commonly cultivated grape, *Vitis vinifera*, is one of the oldest cultivated plants. The first evidence of cultivated vines is found in the Assyrians around 4000 BC. By around 1000 BC, grape plantations were spreading throughout the Mediterranean.

The ancient Greeks and Romans introduced viticulture and winegrowing to Europe. The Romans planted vines throughout France, including the Bordeaux and Bourgogne regions, as well as parts of Germany, Hungary, Austria and even England. Today, grapes are cultivated throughout much of the world, mostly in temperate climates, but also in some subtropical areas.

## France

Generally the best quality wines come from northern districts but the bulk of French wine is a lesser quality, produced in the middle and southern regions. Annual rainfall varies from around 53-75cm (21 inches to 30 inches) in French winegrowing areas.

## Germany

German winegrowing areas have less predictable and overall cooler climates than France. In order to achieve the best results, the Germans have become meticulous in choosing the best available sites (in terms of soil and microclimate). Germany produces some red wine, but mostly white.

## Other Parts of Europe

Wine is also produced in Switzerland, Austria and, in more recent times, southern England and Wales. Prior to World War II, vines were rarely seen outside greenhouses in Britain. Further back in British history, vineyards were in existence throughout southern Britain. The Domesday book of 1085 refers to 38 vineyards.

Britain does pose some particular problems for growing grapes, including lack of heat, uneven rainfall, and high humidity at times; however, there is a growing and very successful viticulture industry in the UK, producing good light wines.

## America



Cortez planted the first vines in Peru in 1524. By 1700, grape plantings had been well established in Southern California and parts of Mexico (generally by missions). Interestingly, the most widely planted grape for a long time in California was the "Mission" grape.

Major winegrowing areas of California today are Sonoma, Napa Valley, Livermore Valley, Contra Costa, Santa Clara and Monterey. Further north along the western coast of America, grapes are now being grown in Oregon, Washington and even parts of British Columbia. Despite a largely unsuitable climate, there are parts of Canada which have been able to support successful, commercial viticulture (including the Niagara Peninsula of Ontario, and the Okanagan valley of British Columbia). Parts of eastern and central USA have also been successfully used for vineyards. The most significant area outside California in the USA is New York State.

## Other Areas

There are many other significant grape growing areas around the world including Australia, New Zealand, South Africa, China, the Middle East, and South America.

### *Some of Australia's Major Winegrowing Areas*

| LOCALITY                         | WINES                    | VARIETIES GROWN                                                                                                                     |
|----------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Adelaide Hills (South Aust.)     | Red and white            | Riesling, Chardonnay, Shiraz, Malec, Sauvignon Blanc, Semillon, Pinot Noir, Cabernet Sauvignon, Chenin Blanc, Gewurtztraminer.      |
| Coonawarra (South Aust.)         | Red and white            | Shiraz, Cabernet Sauvignon, Malbec, Riesling, Chardonnay, Sauvignon Blanc, Pinot Noir, Merlot.                                      |
| Padthaway –Keppoch (South Aust.) | White and red            | Most of those above plus Crouchen, Muscadelle, Muller, Sylvaner and Thurgau.                                                        |
| Yarra Valley (Vic)               | Mainly red               | Cabernet Sauvignon, Pinot Noir, Merlot, Shiraz, Riesling, Sauvignon Blanc, Semillon, Gewurtztraminer.                               |
| Great Western (Vic)              | White, red and sparkling | Chasselas, Ondenc, Pinot Meunier, Chardonnay, Riesling, Merlot, Malbec, Cabernet Sauvignon, Pinot Noir, Shiraz and Gewurtztraminer. |
| Mount Barker (WA)                | Red and white            | Cabernet Sauvignon, Shiraz, Pinot Noir, Malbec, Merlot, Riesling, Chardonnay, Semillon, Sauvignon Blanc and Gewurtztraminer.        |
| Granite Belt, Stanthorpe (Qld)   | White and red            | Chardonnay, Semillon, Sauvignon Blanc, Muscat, Gordo Blanco, Emerald Riesling, Chiraz, Cabernet Sauvignon, Merlot and Malbec.       |

Other important Australian winegrowing areas include the Hunter Valley (north of Sydney), Mornington Peninsula (south of Melbourne), Margaret River (south of Perth) and Tamar Valley (Tasmania); new regions are also being developed in a number of other localities.

## The Grape

Grapes belong to the Vitaceae family. Within this family, only the genus **Vitis** is of any great interest to viticulture, although four of the nine genera in this family yield grapes. The *Vitis* genus includes some 60 to 80 evergreen and deciduous shrubs, mainly of a climbing habit, supporting themselves by tendrils.

Almost all commonly cultivated grapevines belong to the species *Vitis vinifera*, although other species have some use in viticulture: for rootstocks, materials for hybridisation and, in some circumstances, for actual grape production. The commonly grown grape vine (*Vitis vinifera* – also known as the European Grape) originated from Asia Minor and has been carried with civilisation for thousands of years. *V. vinifera* was taken to Mexico by the Spaniards. English settlers took Old World grapes with them and planted them along the Atlantic seaboard. These however failed due to the presence of the insect phylloxera, and fungus diseases like Black Rot, Downy Mildew and Powdery Mildew, as well as deleterious effect of low winter temperatures and hot humid summers.

*V. vinifera* requires a warm temperate climate, with minimum temperatures of -2°C while dormant, -1°C at bud burst and -0.5°C when in full flower. The root system is deep, and as such can draw water from lower levels of the soil; hence the need for high rainfall or irrigation is only moderate. Effective irrigation methods in suitable climates can, however, improve quality and quantity of yields.

The vine does not tolerate wet soils in summer but will tolerate some wetness in winter.

When on a trellis it will tolerate wind reasonably well, but not strong gale force winds. Though sandy soils are preferred, grapes will tolerate most soils provided they are deep and well drained.

Some other grape species that are of significance to viticulture include:

*Vitis amurensis*

*Vitis labrusca*

*Vitis riparia*

*Vitis rupestris*

*Vitis berlandieri*

*Vitis aestivalis*

*Vitis cinerea*

*Vitis rotundifolia*



## The Genera *Vitis* and *Muscadinia*

Plants from both of these genera are called “grapes”. The world viticulture industry concentrates on growing cultivars of *Vitis vinifera*. Cultivars of other species from both genera are however grown for edible fruit in various parts of the world.

*Vitis* has forked tendrils, sheds its bark, has a diaphragm (continuous pith) at the nodes, and elongated clusters with berries that stick to the pedicels at maturity.

*Muscadinia* has tight bark that does not shed, simple tendrils that do not fork, nodes without a diaphragm, and small clusterlets with berries that detach as they mature.

*V. vinifera* also have intermittent tendrils, thin, smooth shiny leaves with 3, 5, or 7 lobes. The berry size varies.

## Species used for Grape Production

*V. vinifera* cultivars produce over 90% of the world's grapes, whether as pure *vinifera* or hybridised. The most important grape species used in North America are:

*V. labrusca* - Concord, Niagara

*V. aestivalis* - Norton, Delaware

*V. vulpina* - Elvira, Clinton

*V. rotundifolia* - Scuppernong, Eden, Muscadine

*V. rupestris* - Rupestris St. George

The Concord variety makes about 80% of the total American production. Many grape varieties have been crossbred between species and even other hybrids to produce improved characteristics of the fruit, growth habit or even disease resistance.

## Root Stocks

Root stocks that exhibit resistance to phylloxera have become invaluable in the industry. Some of the American species used as root stock that have resistance are *Vitis riparia*, *V. berlandieri*, *V. rupestris*, *V. aestivalis*, *V. cordifolia*, and *V. monticola*.

Rootstocks that exhibit resistance to nematodes are also important. These include from America: *Vitus rotundifolia*, *V. champini*, *V. candicans*, and *V. longii*.

## Classification of Grape Varieties

Grapes can be classified based on their use:

### Table Grapes

These are grapes which are sold and used fresh. These varieties must look and taste good, and resist bruising or other damage when handled.

Preferred qualities are:

- Large berries
- Even-sized berries
- Strong skin
- Strong stems
- Good shelf life
- Seedless (in some markets)
- Bunches which are neither loose nor dense (berries should not be too sparse nor too tightly together)

Some varieties used as table grapes include Cardinal, Black Muscat, White Muscat, Waltham Cross, Purple Cornichon, Flame Seedless and Marroo Seedless.

### Wine Grapes

These grapes are crushed and fermented to produce wine. Red or rosé wines are produced by fermenting after crushing while the grape skins are still present. For white wines, the skins are removed before fermentation. Some varieties have skins that add more colour to the wine, others less.

Grapes with high acid content and low sugar will produce dry wines. Grapes with high sugar and lower acid produce sweeter wines. The amount of acid and sugar in a grape depends upon the variety of grape, plant culture, and the stage at which it is picked.

Mechanically harvested wine grapes should have berries which detach easily from the stems and have thick skins which don't damage readily. Thin-skinned grapes must be harvested more carefully.

Varieties commonly used for high quality red wine include Cabernet Sauvignon, Cabernet Franc, Malbec, Merlot, Shiraz and Pinot Noir.

Pinot Noir is often used for rosé.

Varieties often used for quality white wine include Chardonnay, Semillon, Sauvignon Blanc and Gewurztraminer.

### Dried Fruit

Virtually any grapes can be dried, but the varieties used commercially are generally seedless types which ripen fast and at a dry time of year. They must not get moist and split near to harvest. The texture should be soft, and the fruit shouldn't stick together too much in storage. Sultanas (a white seedless variety) are particularly valuable, producing large dried fruits. Raisins may be produced from various varieties, often smaller fruits, including Thomson Seedless, Muscat of Alexandria and Black Corinth.

### Juice Grapes

Unfermented grape juice has gained increased popularity in recent times. Processes (eg. pasteurisation) used to preserve the juice can have a detrimental effect on the flavour with some grape varieties, while others are not dramatically altered. Several varieties may be blended to produce juice.

### **Canned Grapes**

Seedless grapes are sometimes canned, either alone or with other fruit as fruit salad. Thompson Seedless is commonly used for this purpose in the USA.

## **Vineyard Mailing Lists**

Many vineyards around the world operate mailing lists. They post out newsletters or other information either in the mail or over the internet. Some run wine clubs that sell wine through the mail.

It is a valuable exercise to join a few of these mailing lists to (amongst other things) see how these vineyards use mail lists as an effective marketing tool.

